Evaluation of a Novel Large Language Model (LLM)-powered Chatbot for Oral Boards Scenarios

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Abstract

While previous studies have demonstrated that generative artificial intelligence (AI) can pass medical licensing exams, AI's role as an examiner in complex, interactive assessments remains unknown. AI-powered chatbots could serve as educational tools to simulate oral examination dialogues. Here, we present initial validity evidence for an AI-powered chatbot designed for general surgery residents to prepare for the American Board of Surgery (ABS) Certifying Exam (CE). We developed a chatbot using GPT-4 to simulate oral board scenarios. Scenarios were completed by general surgery residents from six different institutions. Two experienced surgeons evaluated the chatbot across five domains: inappropriate content, missing content, likelihood of harm, extent of harm, and hallucinations. We measured inter-rater reliability to determine evaluation consistency. Seventeen residents completed a total of 20 scenarios. Commonly tested topics included small bowel obstruction (30%), diverticulitis (20%), and breast disease (15%). Based on two independent reviewers, evaluation revealed 11-25% of chatbot simulations had no errors and an additional 11%-35% contained errors of minimal clinical significance. The chatbot limitations included incorrect management advice and critical omissions of information. This study demonstrates the potential of an AI-powered chatbot in enhancing surgical education

through oral board simulations. Despite challenges in accuracy and safety, the chatbot offers a novel approach to medical education, underscoring the need for further refinement and standardized evaluation frameworks. Incorporating domain-specific knowledge and expert insights is crucial for improving the efficacy of AI tools in medical education.